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UNIVERSITY OF APPLIED SCIENCES UPPER AUSTRIA

ANGIE
ACADEMIC NETWORK FOR A GREEN & INNOVATIVE EUROPE

WP2–Setting-up the Graduate Study Program (GSP)
"Transition to Green for Sustainable Society"

Academic Network for a Green and Innovative Europe

D2.2 – Defining the competences to be acquired by the students
D2.3 – Defining the main structure of the new GSP:

- *Curriculum / Chapters*
- *Number of contact hours*
- *Credits / Micro-credits*

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1 Foreword

After identifying the needs/deficiencies in the society, in the WATERIC field (D2.1.1), and analysing the results of that study, a set of disciplines were defined based on the most relevant topics (D2.1.2). With the disciplines/ curricular units (CU) defined it is necessary to create for each CU the require competencies and to define the learning outcomes that will lead to the acquisition of those competencies. Finally, the curriculum structure of the entire master needs to be designed.

2 Overview

The main objective of these deliverables is the design-creation-functionalization of a new program study, in a harmonised design. Considering the subjects/CU already defined in D2.1.2, this document pretends to show the learning outcomes associated to each UC and the study program/curriculum developed considering the new graduate study program. When creating this program, it was considered the existing national master´s, in the different partners countries, with the aim of creating a harmonize master study program.
Additionally, and considering the information require to create the new study program, for each CU, a form was created whose template can be observed in figure 1. This form has the most relevant information relatively to the discipline, namely, objectives, learning outcomes, skills, knowledge, Transversal competences, content, nº of credits, Learning Method(s) and Delivery Mode(s), and will be created one for each UC in the WATERIC field.

UC name			ECTS:
	Please give details below		
Characteristics of the discipline	Discipline type Compulsory /optional Optional	Discipline Pre-requisite	Assessment summary
Full details of assessment and respective weight			
Learning Method(s) and Delivery Mode(s)			
Main Aim			
Objectives			
Learning Outcomes			
Knowledge			
Skills			
Transverse / soft competencies			
Contents			

Figure 1- Example of the Curricular Unit Form

3 Curricular Unit form

In this section a detailed information of the disciplines selected, in the WATERIC filed, to be part of the new master program study will be presented, by activity sector, for an easier visualization.

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3.1 Construction-Industrial activities disciplines

3.1.1 Critical raw Materials

Critical Raw Materials			ECTS: 7
Characteristics of the discipline	O.I.#02.1/1-1		
	Discipline type Compulsory /optional Optional	Discipline Pre-requisite No pre-requisites	Assessment summary Minimum Mark: 9.5 / 20
Full details of assessment and respective weight	Exam: 2h, + project, PBL, etc.		
Learning Method(s) and Delivery Mode(s)	According to D.2.5 and D.2.4		
Main Aim	The primary aim of the "Critical Raw Materials" discipline is to equip students with a comprehensive understanding of the importance, challenges, and management strategies related to critical raw materials (CRMs) essential for modern technologies and sustainable development. The course seeks to provide in-depth knowledge of the environmental, economic, and geopolitical factors that affect the supply and utilization of CRMs and to develop the skills necessary to propose innovative and sustainable solutions for managing these resources effectively. The discipline emphasizes a holistic approach, integrating technical, strategic, and ethical considerations in the responsible use and management of CRMs.		
Objectives	• Grasp the concept of critical raw materials, their global distribution, and the factors influencing their criticality, including geopolitical, environmental, and economic aspects. • Assess the environmental and social challenges of CRM extraction, use, and disposal, and understand strategies to mitigate these impacts. • Formulate innovative solutions for reducing dependency on CRMs, such as material substitution, recycling, and efficient resource management, while ensuring economic and environmental viability.		
Learning Outcomes	• Explain the concept of critical raw materials (CRMs) and why they are essential for industries such as renewable energy, electronics, aerospace, and automotive sectors. • Identify the main factors that contribute to the criticality of raw materials, including scarcity, supply chain risks, and geopolitical considerations. • Understand the global distribution of CRMs and the geopolitical challenges that affect their supply and access.		

- Analyse case studies of supply disruptions and their impact on global industries and economies.
- Describe the environmental impact of extracting, processing, and using CRMs, including pollution, biodiversity loss, and greenhouse gas emissions.
- Discuss the social issues associated with CRM mining, such as labour conditions, community displacement, and ethical sourcing.
- Demonstrate knowledge of the current technologies and strategies used for substituting CRMs with more sustainable or abundant alternatives.
- Explain the challenges and opportunities associated with the recycling and recovery of CRMs from end-of-life products.
- Familiarize with international and regional policies related to CRM management, including EU strategies and initiatives like the European Raw Materials Alliance.
- Understand regulatory frameworks that promote the responsible sourcing and sustainable use of CRMs.
- Conduct assessments of raw material supply risks and develop strategies to mitigate these risks for specific industries or products.
- Use analytical tools to model and predict future trends in the availability and demand for CRMs.
- Design and propose efficient resource management practices that reduce dependency on CRMs and enhance material efficiency.
- Develop strategies for implementing circular economy principles in industrial processes to minimize waste and maximize material reuse.
- Analyse market data to understand trends in CRM supply and demand, price fluctuations, and potential disruptions.
- Conduct research on new technologies for CRM extraction, processing, and recycling, and evaluate their feasibility and impact.
- Apply knowledge of material science to research and suggest viable alternatives for critical raw materials, assessing trade-offs in performance, cost, and environmental impact.
- Develop and test innovative methods for enhancing the recyclability and recovery of CRMs from complex products.
- Formulate policy recommendations to governments or organizations for improving CRM supply security and promoting sustainable practices.
- Work collaboratively on developing strategic plans for industries to reduce CRM dependency while maintaining product performance.

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	• Cultivate a deep appreciation for sustainable resource management and the importance of balancing economic, environmental, and social considerations in the use of CRMs. • Adopt a proactive approach to minimizing the environmental impact of raw material extraction and promoting responsible consumption. • Embrace a solution-oriented attitude when facing the challenges of CRM scarcity, seeking innovative and interdisciplinary approaches to address them. • Demonstrate a willingness to explore and implement new technologies and business models that enhance CRM sustainability.
Knowledge	• Understanding the geopolitical factors influencing the availability and accessibility of critical raw materials. • Knowledge of environmentally friendly and sustainable mining techniques. • Skills to ensure traceability and transparency in the supply chain to improve accountability and sustainability. • Knowledge of advanced recycling technologies and their application to critical raw materials • Understanding and adhering to international, national, and local regulations concerning critical raw materials.
Skills	• Skills to conduct risk assessments related to the supply and demand of critical raw materials. • Ability to develop and apply waste management practices to minimize environmental impact. • Ability to designing and implement efficient supply chain strategies for critical raw materials. • Ability to apply principles of the circular economy to enhance material reuse and recycling. • Skills to contribute to the development of policies aimed at securing critical raw material supply.
Transverse / soft competencies	• Critical thinking for the evaluation of complex scenarios involving critical raw materials to take informed decisions • Capacity to solve various problems when address challenges in the supply, extraction, and processing of critical raw materials. • Innovating approach to develop solutions for the sustainable use and management of critical raw materials. • Analytical skills to be able to analyse data and trends related to critical raw materials to inform strategy and decision-making. • Capacity of effectively communication to provide with complex information regarding critical raw materials to diverse stakeholders. • Capacity to collaborate with team members, in multidisciplinary teams, in order to address issues related to critical raw materials.

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	• Capacity to lead initiatives to promote sustainable practices and advocate for policies supporting critical raw materials. • Ensuring ethical considerations are central to all decisions and practices involving critical raw materials.
Contents	• Identification and assessment of critical raw materials • Sustainable extraction and processing techniques • Substitution and recycling of critical raw materials • Supply chain management and security in raw materials • Policy and regulatory frameworks

Code	Title	Number of theoretical contact hours	Number of practical contact hours	Number of individual learning hours	ECTSs and µECTSs
O.I.#02.1/1-1	Critical Raw Materials	28	56	126	7
Type: Optional	Identification and assessment of critical raw materials	6	12	26	1,5
Domain: Industrial	Sustainable extraction and processing techniques	6	12	25	1,4
Discipline: #02	Substitution and recycling of critical raw materials	6	12	25	1,4
Year: 1	Supply chain management and security in raw materials	6	12	25	1,4
Semester: 1	Policy and regulatory frameworks	4	8	25	1,2
Hours	210	28	56	126	7
ECTSs	7				7,0

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3.1.2 Resources Management

Resources Management			ECTS: 7
Characteristics of the discipline	O.I.#02.2/1-1		
	Discipline type Compulsory /optional Optional	Discipline Pre-requisite No pre-requisites	Assessment summary Minimum Mark: 9.5 / 20
Full details of assessment and respective weight	Exam: 2h, + project, PBL, etc.		
Learning Method(s) and Delivery Mode(s)	According to D.2.5 and D.2.4		
Main Aim	The primary aim of the "Resources Management" discipline is to develop students' ability to manage natural and human-made resources sustainably in engineering and industrial contexts. The course focuses on strategic planning, efficient allocation, and optimization of resources to minimize environmental impact while maximizing economic and social benefits. The discipline prepares students to address complex challenges associated with resource scarcity, environmental degradation, and sustainability, using innovative and data-driven approaches.		
Objectives	• Understand the theories and practices of resource efficiency, conservation, and sustainable use in engineering and industrial contexts. • Use quantitative tools like energy audits and material flow analysis to assess and improve resource use and environmental impact. • Develop and implement strategic plans for optimizing resource use, minimizing waste, and managing supply chains sustainably.		
Learning Outcomes	• Develop and implement resource management strategies that optimize resource use while ensuring sustainability and cost-effectiveness. • Use resource planning tools and software to model and analyse resource flows, consumption patterns, and environmental impacts. • Apply quantitative methods, such as material flow analysis (MFA) and energy audits, to assess the efficiency of resource use and identify areas for improvement. • Perform life cycle costing (LCC) and economic analysis to evaluate the financial viability of resource-saving initiatives.		

	• Plan, execute, and monitor resource management projects, ensuring that they align with sustainability goals and regulatory requirements. • Manage supply chains with a focus on resource efficiency, considering logistics, procurement, and waste minimization strategies. • Conduct risk assessments related to resource availability, environmental impact, and regulatory compliance, and develop mitigation strategies. • Use decision-making frameworks to balance trade-offs between resource efficiency, economic performance, and environmental protection. • Prepare and present resource management plans and impact assessments to diverse audiences, including industry stakeholders, policymakers, and community members. • Engage with stakeholders to build consensus on resource management strategies and foster collaboration for sustainable solutions. • Develop a strong commitment to resource stewardship, prioritizing long-term sustainability over short-term gains. • Demonstrate a personal and professional responsibility for minimizing environmental impact and promoting resource conservation. • Uphold high ethical standards when managing resources, ensuring transparency and fairness in resource allocation and use. • Advocate for responsible resource management that respects the rights and needs of future generations. • Cultivate a proactive attitude toward identifying and implementing innovative solutions for resource optimization and environmental protection. • Embrace continuous learning and adapt to new technologies and practices that enhance resource management. • Value interdisciplinary collaboration and work effectively with professionals from different backgrounds to develop integrated resource management strategies. • Appreciate the importance of engaging with local communities and understanding their perspectives on resource use and sustainability.
Knowledge	• Understanding the foundational concepts of sustainability, including the Triple Bottom Line (environmental, social, economic impacts). • Familiarity with sustainability standards, such as ISO 14001 (Environmental Management Systems) and LEED (Leadership in Energy and Environmental Design). • Knowledge of how to assess and manage natural resources (water, minerals, energy) sustainably.

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	• Concepts of circular economy, life cycle assessment (LCA), and materials flow analysis. • Strategies for waste reduction and recycling within engineering processes. • Methods for evaluating the potential environmental consequences of engineering projects.
Skills	• Competence in designing and managing projects that prioritize resource efficiency and sustainability. • Proficiency in conducting LCA to evaluate the environmental performance of engineering products and systems. • Skills in presenting complex sustainability concepts in clear, understandable language. • Engineering design skills focused on minimizing resource use and maximizing environmental performance. • Proficiency in sustainable materials selection and eco-design principles. • Applying creative and innovative thinking to develop new, sustainable solutions in engineering projects. • Ability to adapt and optimize existing systems to improve resource efficiency.
Transverse / soft competencies	• A genuine passion for advancing sustainability in engineering practices, and willingness to advocate for sustainable practices within teams and organizations • Recognizing the ethical implications of resource management decisions on the environment, and society and commitment to transparency and integrity in reporting and decision-making. • Openness to continually updating knowledge and skills as technologies and regulations evolve, and a flexible mindset that embraces new methods and approaches to sustainable engineering challenges. • An appreciation for working in diverse, multidisciplinary teams to address complex sustainability issues. • Sensitivity to the cultural and environmental contexts in which engineering projects take place, and an understanding of how resource management practices impact communities and ecosystems globally. • A hands-on, proactive approach to solving resource management challenges, and willingness to take initiative in developing and implementing sustainable engineering solutions
Contents	• Foundations of sustainable resource management • Life Cycle Assessment (LCA) and Environmental Impact Analysis (EIA) • Circular economy in engineering • Sustainable energy and resource use • Economic and policy aspects of sustainability • Risk management and climate change adaptation • Systems thinking and modelling for resource management • Technological innovations for sustainable engineering • Ethical and social dimensions of sustainability

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	• Practical applications and case studies

Code	Title	Number of theoretical contact hours	Number of practical contact hours	Number of individual learning hours	ECTSs and µECTSs
O.I.#02.2/1-1	Resources Management	28	56	126	7
Type: Optional	Foundations of sustainable resource management	2	4	9	0,5
Domain: Industrial	Life Cycle Assessment (LCA) and Environmental Impact Analysis (EIA)	2	6	13	0,6
Discipline: #02	Circular economy in engineering	2	6	13	0,7
Year: 1	Sustainable energy and resource use	2	4	9	0,5
Semester: 1	Economic and policy aspects of sustainability	2	4	9	0,5
	Risk management and climate change adaptation	2	4	9	0,5
	Systems thinking and modelling for resource management	2	4	9	0,5
	Technological innovations for sustainable engineering	4	4	11	0,6
	Ethical and social dimensions of sustainability	2	4	9	0,5
	Practical applications and case studies	8	16	39	2,1
Hours	210	28	56	126	7
ECTSs	7				7,0

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3.1.3 Materials in Circular Economy

Materials in Circular Economy			ECTS: 7
Characteristics of the discipline	O.I.#02.3/1-1		
	Discipline type Compulsory /optional Optional	Discipline Pre-requisite No pre-requisites	Assessment summary Minimum Mark: 9.5 / 20
Full details of assessment and respective weight	Exam: 2h, + project, PBL, etc.		
Learning Method(s) and Delivery Mode(s)	According to D.2.5 and D.2.4		
Aim	The primary aim of the "Materials in Circular Economy" discipline is to provide students with a deep understanding of the principles and practices of the circular economy as they pertain to material management. The course seeks to develop knowledge and skills to design, utilize, and manage materials in ways that minimize waste, extend product lifecycles, and promote sustainable resource use. It aims to prepare students to innovate and implement circular strategies in various sectors, ensuring environmental sustainability, economic viability, and social responsibility.		
Objectives	• Comprehend the principles and significance of the circular economy, especially regarding sustainable material management. • Develop practical and innovative material solutions that support recycling, reuse, and remanufacturing, aligned with economic and environmental objectives. • Use analytical tools to assess material flows and propose strategies for material efficiency, waste reduction, and closed-loop systems.		
Learning Outcomes	• Demonstrate a comprehensive understanding of circular economy principles and their application to material design and management. • Analyse the environmental impacts of different materials using life cycle assessment (LCA) and apply findings to minimize ecological footprints. • Explain the processes involved in recycling and upcycling various materials, including metals, polymers, and composites, and evaluate their feasibility in industrial settings. • Assess the role of emerging materials and technologies in enabling circularity and sustainable resource use.		

	• Apply tools and methodologies, such as LCA and material flow analysis, to optimize material use and reduce waste. • Design innovative, circular product solutions that prioritize recyclability, reuse, and minimal environmental impact. • Manage projects related to sustainable material usage, including planning, execution, and monitoring of circular economy initiatives. • Conduct critical assessments of material supply chains and develop strategies to implement closed-loop systems effectively. • Exhibit a collaborative spirit, valuing interdisciplinary teamwork and stakeholder engagement in developing circular material solutions.
Knowledge	• Understanding the core concepts of the circular economy, such as material loops, waste minimization, resource efficiency, and regenerative systems. • Understanding the environmental impact of materials throughout their life cycle. • Techniques to assess the sustainability of materials and products, focusing on their life cycle from extraction and manufacturing to disposal or recycling. • Knowledge of managing wastes and the recycling processes for various types of materials (e.g., metal recycling, plastic upcycling) • Principles of designing products and systems that facilitate material reuse, disassembly, and recycling.
Skills	• Skills to analyse and model the flow of materials in industrial systems, identifying opportunities for waste reduction and resource recovery, and use of tools and software for material flow analysis • Proficiency in conducting LCAs to measure the environmental impact of materials and design strategies to minimize it • Analysing data on material use, waste streams, and recycling rates to inform strategic decisions • Skills to interpret and apply regulations related to waste management and material use
Transverse / soft competencies	• A genuine commitment to reducing environmental impact through sustainable materials and circular practices, and eagerness to innovate and improve current material management systems. • Ability to understand and approach problems with a holistic perspective, considering how different components of a system interact. • Being receptive to new ideas, technologies, and methods that enhance material circularity, and willingness to adapt strategies as new information and innovations become available.

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	• Taking initiative in finding creative and practical solutions to material-related challenges. • Understanding the ethical implications of material use and waste, and striving for responsible and transparent decision-making. • Appreciating the importance of teamwork and collaboration across disciplines and industries to implement circular solutions.
Content	• Introduction to Circular Economy Concepts • Material Properties and Their Impact on Circularity • Design Strategies for Circular Products • Material Flow Analysis and Waste Management • Sustainable Material Innovation • Advanced Manufacturing for the Circular Economy • Regulations, Standards, and Compliance

Code	Title	Number of theoretical contact hours	Number of practical contact hours	Number of individual learning hours	ECTSs and µECTSs
O.I.#02.3/1-1	Materials in Circular Economy	28	56	126	7
Type: Optional	Introduction to Circular Economy Concepts	4	8	19	1,0
Domain: Industrial	Material Properties and Their Impact on Circularity	5	10	24	1,2
Discipline: #02	Design Strategies for Circular Products	4	8	19	1,0
Year: 1	Material Flow Analysis and Waste Management	4	8	19	1,0
Semester: 1	Sustainable Material Innovation	4	8	19	1,0
	Advanced Manufacturing for the Circular Economy	6	12	29	1,5
	Regulations, Standards, and Compliance	1	2	4	0,2
Hours	210	28	56	126	7
ECTSs	7				7,0

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3.1.4 Green Buildings (nZEB)

Green Buildings (nZEB)			ECTS: 5
Characteristics of the discipline	C.C.#13/2-3		
	Discipline type Compulsory /optional Compulsory	Discipline Pre-requisite No pre-requisites	Assessment summary Minimum Mark: 9.5 / 20
Full details of assessment and respective weight	Exam: 2h, + project, PBL, etc.		
Learning Method(s) and Delivery Mode(s)	According to D.2.5 and D.2.4		
Main Aim	The primary aim of the "Green Buildings (nZEB)" discipline is to equip students with the knowledge and skills needed to design, construct, and manage energy-efficient buildings that adhere to the nearly Zero Energy Building (nZEB) standards. The course focuses on sustainable building practices, advanced energy systems, and the integration of renewable energy technologies to minimize energy consumption and environmental impact. It prepares students to create built environments that are both ecologically responsible and economically viable, contributing to global efforts in reducing carbon emissions and promoting sustainable urban development.		
Objectives	• Learn the key principles and requirements for nearly Zero Energy Buildings, including energy-efficient design and renewable energy integration. • Conduct energy modelling and simulations to enhance building performance, focusing on efficient building envelopes and passive design strategies. • Design and manage green building projects, incorporating renewable energy systems, sustainable materials, and smart technologies to achieve nZEB performance.		
Learning Outcomes	• Explain the core principles of nearly Zero Energy Buildings (nZEB) and the role of green buildings in achieving sustainability goals. • Understand the technical requirements and design considerations for energy efficiency, renewable energy integration, and high-performance building envelopes. • Evaluate different renewable energy technologies and their suitability for various building types and geographic locations.		

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	• Describe the impact of building materials and construction practices on energy efficiency and indoor environmental quality (IEQ). • Utilize energy modelling software to simulate and optimize building energy performance to meet nZEB standards. • Apply passive and active design strategies to minimize energy use while ensuring occupant comfort and well-being. • Develop and implement smart building solutions that use automation and data-driven technologies for enhanced energy management. • Prepare comprehensive documentation for green building certifications and ensure compliance with relevant regulations and standards. • Embrace a commitment to sustainability, focusing on minimizing the environmental impact of the built environment through responsible design and construction practices. • Demonstrate an ethical and holistic approach to building design, prioritizing occupant health and environmental quality. • Cultivate a willingness to learn and adapt to new technologies, materials, and design methodologies that enhance building sustainability. • Appreciate the value of collaboration, actively working with stakeholders to create integrated and sustainable building solutions.
Knowledge	• Knowledge of green building certifications and standards • Understanding of net-zero energy and net-zero water concepts in building design and operation • Competence in conducting energy audits and energy modelling for buildings • Knowledge of water-efficient fixtures, rainwater harvesting systems, and greywater reuse technologies • Knowledge of IAQ testing protocols, green materials selection, and maintenance practices to enhance indoor environmental quality
Skills	• Ability to interpret and apply criteria for sustainable site development, water efficiency, energy efficiency, materials selection, and indoor environmental quality • Ability to design and implement strategies to achieve net-zero energy consumption and water use • Skills in optimizing building envelope design, HVAC systems, lighting, and appliances for energy efficiency • Ability to calculate water demand, assess water efficiency measures, and optimize water use in buildings

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	• Understanding of factors influencing indoor air quality (IAQ) such as ventilation, filtration, and material emissions
Transverse / soft competencies	• Analytical skills to evaluate building performance metrics and identify opportunities for improvement. • Ability to troubleshoot and resolve complex challenges related to green building design and operation. • Capacity to integrate sustainable design principles into architectural and engineering solutions • Effective communication of green building strategies and benefits to diverse stakeholders (e.g., clients, designers, contractors). • Ability to convey technical information clearly and persuasively. • Ability to innovate and explore emerging trends in green building design and construction • Understanding of ethical considerations in sustainable building practices (e.g., social equity, environmental justice).
Contents	• Green Building Standards • Near Zero Emission Buildings • Energy Performance • Water Conservation • Indoor Environmental Quality

Code	Title	Number of theoretical contact hours	Number of practical contact hours	Number of individual learning hours	ECTSs and µECTSs
C.C.#13/2-3	Green Buildings (nZEB)	14	28	108	5
Type: Compulsory	Green Building Standards	3	6	24	1,1
Domain: Construction	Near Zero Emission Buildings	3	6	24	1,1
Discipline: #13	Energy Performance	3	6	24	1,0
Year: 2	Water Conservation	3	6	24	1,0
Semester: 3	Indoor Environmental Quality	2	4	18	0,8
Hours	150	14	28	108	5
ECTSs	5				5,0

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3.1.5 LCA and sustainable rehabilitation of buildings

LCA and sustainable rehabilitation of buildings			ECTS: 3
Characteristics of the discipline	C.C.#07/1-2		
	Discipline type Compulsory /optional	Discipline Pre-requisite	Assessment summary
	Compulsory	No pre-requisites	Minimum Mark: 9.5 / 20
Full details of assessment and respective weight	Exam: 2h, + project, PBL, etc.		
Learning Method(s) and Delivery Mode(s)	According to D.2.5 and D.2.4		
Main Aim	The main aim of the "LCA and Sustainable Rehabilitation of Buildings" discipline is to provide students with the knowledge and skills to evaluate and minimize the environmental impact of buildings through sustainable rehabilitation practices. The course emphasizes the use of Life Cycle Assessment (LCA) as a critical tool for assessing the energy and resource efficiency of buildings, guiding students in designing effective strategies for retrofitting and improving the sustainability of existing structures. The ultimate goal is to prepare students to contribute to the development of energy-efficient, low-carbon, and environmentally responsible built environments.		
Objectives	• Understand and apply life cycle assessment (LCA) techniques to evaluate the environmental impacts of buildings over their entire lifecycle. • Develop strategies for the sustainable rehabilitation and retrofitting of buildings, enhancing energy efficiency and reducing environmental footprints. • Use assessment tools and technologies to monitor and improve the energy performance of existing buildings during and after rehabilitation.		
Learning Outcomes	• Explain the fundamental principles and methodology of Life Cycle Assessment (LCA) and its application in assessing the environmental impacts of building materials and construction processes. • Identify the various stages of a building's life cycle, from material extraction and construction to use, maintenance, and end-of-life, and understand how this impact the environment. • Demonstrate knowledge of sustainable rehabilitation techniques and strategies that can be applied to existing buildings to improve their energy performance, resource efficiency, and overall sustainability. • Understand the criteria for selecting environmentally friendly materials and construction methods that minimize waste and carbon emissions during rehabilitation.		

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- Familiarize with international and regional regulations, standards, and guidelines for building rehabilitation and the use of LCA in sustainable construction projects.
- Comprehend the role of sustainable rehabilitation in urban planning and its alignment with sustainability goals, such as those outlined in the European Green Deal.
- Understand diagnostic methods used to assess the condition of existing buildings, including structural health monitoring and energy performance analysis, to inform rehabilitation strategies.
- Analyse case studies of successful sustainable building rehabilitation projects to identify best practices and lessons learned.
- Conduct comprehensive Life Cycle Assessments for building rehabilitation projects using industry-standard tools (such as GaBi, SimaPro, or OpenLCA) to evaluate environmental impacts and identify improvement opportunities.
- Interpret LCA results to make data-driven recommendations for material selection, construction methods, and energy retrofitting.
- Design and propose energy-efficient retrofitting solutions, such as improved insulation, energy-efficient windows, and HVAC system upgrades, for existing buildings.
- Apply knowledge of renewable energy technologies to suggest viable energy solutions, such as solar panels or geothermal heating, in rehabilitation projects.
- Manage sustainable rehabilitation projects by developing plans that include timelines, budgets, and resource allocation, while ensuring compliance with environmental regulations.
- Use digital tools like Building Information Modelling (BIM) to simulate and optimize building performance before implementing rehabilitation measures.
- Perform risk assessments to identify potential environmental, structural, or economic challenges associated with building rehabilitation.
- Develop strategies to mitigate identified risks, ensuring the sustainability and longevity of rehabilitation efforts.
- Prepare clear and comprehensive reports that communicate LCA findings and sustainable rehabilitation strategies to stakeholders, including building owners, architects, and regulatory bodies.
- Facilitate discussions with stakeholders to align rehabilitation efforts with community needs and sustainability goals.
- Cultivate a strong sense of environmental responsibility and a commitment to promoting sustainable practices in building rehabilitation.

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	• Prioritize long-term environmental benefits over short-term gains when making decisions related to the renovation and maintenance of buildings. • Approach challenges with creativity and a focus on continuous improvement. • Demonstrate the ability to work effectively in multidisciplinary teams, respecting the expertise and perspectives of professionals from different fields.
Knowledge	• Knowledge of sustainable materials and innovating green technologies, by understanding their characteristics and potential to be used in building rehabilitation • Understanding and applying principles of energy-efficient heating, ventilation, and air conditioning systems. • Ability to conduct water usage audits to identify and address inefficiencies. • Ability to improve the acoustic performance of rehabilitated buildings. • Understanding and applying standards for green building certifications such as LEED or BREEAM.
Skills	• Ability to integrate sustainable design principles into building rehabilitation projects. • Ability to use software tools to model and simulate the energy performance and to implement the results in the design of automation systems to optimize energy use. • Ability to implement systems to improve indoor air quality. • Ability to enhance the operational efficiency of rehabilitated buildings.
Transverse / soft competencies	• Proving critical thinking by evaluating the sustainability of different materials, technologies, and systems applied in building rehabilitation. • Capability to address and resolve challenges encountered during the sustainable rehabilitation of buildings. • Innovation: Developing innovative and effective solutions for enhancing the sustainability of rehabilitated buildings. • Proving analytical skills by analysing and interpreting data to make informed decisions on sustainable building practices. • Applying effective communication on complex sustainability concepts and benefits to stakeholders. • Working collaboratively with diverse professionals including architects, engineers, and sustainability experts. • Managing sustainable rehabilitation projects to ensure they are completed on time and within budget. • Understanding and applying relevant regulations and standards in sustainable building rehabilitation. • Leading sustainability initiatives and advocating for sustainable practices in building rehabilitation projects.

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	• Upholding ethical standards in all aspects of sustainable building rehabilitation, ensuring social and environmental responsibility.
Contents	• Life cycle Analysis (LCA) • Sustainable materials and technologies for building rehabilitation • Energy-efficient building systems • Water management in rehabilitated buildings • Indoor environmental quality (IEQ) improvement • Sustainable Building Operations and Maintenance

Code	Title	Number of theoretical contact hours	Number of practical contact hours	Number of individual learning hours	ECTSs and µECTSs
C.C.#07/1-2	LCA and sustainable rehabilitation of buildings	14	28	42	3
Type: Compulsory	Life cycle Analysis (LCA)	2	4	7	0,4
Domain: Construction	Sustainable materials and technologies for building rehabilitation	3	6	12	0,6
Discipline: #07	Energy-efficient building systems	3	5	10	0,6
Year: 1	Water management in rehabilitated buildings	2	5	9	0,5
Semester: 2	Indoor environmental quality (IEQ) improvement	2	4	7	0,4
	Sustainable Building Operations and Maintenance	2	4	7	0,4
Hours	90	14	28	48	3
ECTSs	3				3,0

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3.2 Forms for Transportation and Logistics disciplines

3.2.1 Sustainable Supply Chain Management

Sustainable Supply Chain Management			ECTS: 6
Characteristics of the discipline	C.T.#03/1-1		
	Discipline type Compulsory /optional Compulsory	Discipline Pre-requisite No pre-requisites	Assessment summary Minimum Mark: 9.5 / 20
Full details of assessment and respective weight	Exam: 2h, + project, PBL, etc.		
Learning Method(s) and Delivery Mode(s)	According to D.2.5 and D.2.4		
Main Aim	The aim of the "Sustainable Supply Chain Management" discipline is to provide students with comprehensive knowledge and practical skills to design, manage, and optimize supply chains in a way that enhances economic, environmental, and social sustainability. The course focuses on integrating sustainable practices into all elements of the supply chain, from sourcing and production to distribution and end-of-life management, preparing students to contribute to responsible and resilient supply chain solutions.		
Objectives	• To explore the concept of sustainability within supply chain operations and its impact on global business practices, emphasizing environmental and social responsibility. • To identify and evaluate the challenges, risks, and opportunities involved in managing sustainable supply chains, including economic pressures, resource constraints, and environmental regulations. • To create and implement strategic approaches to optimize supply chain performance while ensuring the efficient use of resources and minimizing negative environmental impacts. • To understand how emerging technologies (e.g., blockchain, IoT, AI) can enhance supply chain transparency, traceability, and sustainability. • To foster an understanding of ethical considerations and social impact in supply chain management, promoting practices that improve the well-being of communities and ensure fair labour conditions.		
Learning Outcomes	• Demonstrate an understanding of the principles of sustainable supply chain management and the role it plays in global sustainability. • Identify and explain the environmental, social, and economic impacts of supply chain activities.		

	• Understand regulatory frameworks, standards, and best practices that promote supply chain sustainability. • Apply analytical tools and techniques to assess the environmental and social impacts of supply chains and suggest improvements • Develop strategies to integrate sustainability into supply chain operations, including sustainable procurement, waste reduction, and energy efficiency. • Use data-driven approaches to evaluate supply chain performance and optimize processes for reduced environmental impact. • Exhibit leadership in driving sustainable initiatives within an organization, working effectively in diverse and interdisciplinary teams. • Demonstrate critical thinking and problem-solving skills to address complex supply chain challenges, with a focus on long-term sustainability. • Communicate and collaborate effectively with stakeholders to promote sustainable practices across the supply chain.
Knowledge	• Principles and frameworks of sustainable supply chain management. • Environmental impact assessment methods and life cycle thinking in supply chains. • Global standards and regulations, such as ISO 14001, GRI standards, and the UN Global Compact. • Awareness of technologies that support sustainability, such as digital twins, big data analytics, and supply chain management software. • Understanding of how circular economy principles can be incorporated into supply chains.
Skills	• Ability to conduct supply chain analysis using various tools and techniques to optimize for sustainability. • Skills in developing and implementing strategic plans that address environmental and social responsibilities. • Competence in managing sustainable supply chain projects, including planning, execution, monitoring, and risk management. • Ability to make informed decisions that balance economic performance with sustainability goals. • Proficiency in using digital platforms and software for supply chain analysis and management, with a focus on sustainability metrics.
Transverse / soft competencies	• Ability to lead and inspire teams to adopt sustainable practices and collaborate across departments and industries. • Effective communication skills to engage with various stakeholders, including suppliers, partners, and customers, about sustainability efforts.

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	• A strong sense of ethics and social responsibility, with a commitment to making positive environmental and societal impacts. • Flexibility to adapt to rapidly changing supply chain environments and the ability to innovate in developing sustainable solutions. • Ability to think critically and creatively in solving complex problems related to supply chain sustainability, considering multiple perspectives
Contents	• Integration of Circular Economy Principles in Supply Chain Management • Green Procurement and Supplier Collaboration • Carbon Footprint Reduction in Supply Chain Operations • Ethical and Social Responsibility in Supply Chains • Technology and Innovation for Sustainable Supply Chains

Code	Title	Number of theoretical contact hours	Number of practical contact hours	Number of individual learning hours	ECTSs and µECTSs
C.T.#03/1-1	Sustainable Supply Chain Management	28	28	124	6
Type: Compulsory	Integration of Circular Economy principles in Supply Chain Management	6	6	26	1,3
Domain: Transport	Green procurement and supplier collaboration	6	6	26	1,3
Discipline: #03	Carbon footprint reduction in Supply Chain operations	6	6	26	1,3
Year: 1	Ethical and social responsibility in Supply Chains	4	4	18	0,9
Semester: 1	Technology and innovation for sustainable Supply Chains	6	6	28	1,3
Hours	180	28	28	124	6
ECTSs	6				6,0

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3.2.2 Sustainable Transportation

Sustainable Transportation			ECTS: 7
Characteristics of the discipline	C.T.#09/1-2		
	Discipline type Compulsory /optional Compulsory	Discipline Pre-requisite No pre-requisites	Assessment summary Minimum Mark: 9.5 / 20
Full details of assessment and respective weight	Exam: 2h, + project, PBL, etc.		
Learning Method(s) and Delivery Mode(s)	According to D.2.5 and D.2.4		
Main Aim	The aim of the "Sustainable Transportation" discipline is to equip students with the knowledge and skills to design, analyse, and implement transportation systems that are efficient, environmentally friendly, and socially responsible. The course emphasizes reducing greenhouse gas emissions, minimizing resource consumption, and enhancing mobility in urban and rural settings while considering economic feasibility and community impact.		
Objectives	• To provide an understanding of the principles and challenges of sustainable transportation and their role in mitigating climate change and improving quality of life. • To assess the environmental, economic, and social impacts of different transportation modes and infrastructure projects. • To create innovative transportation strategies that reduce emissions, promote the use of clean energy, and improve accessibility. • To understand the influence of policy, urban planning, and human behaviour on transportation systems and how to drive positive change. • To explore how advancements like electric vehicles (EVs), autonomous systems, and smart transportation infrastructure can contribute to sustainability.		
Learning Outcomes	• Demonstrate an understanding of sustainable transportation principles and their impact on urban development and environmental health. • Explain the benefits and limitations of different transportation technologies and systems from a sustainability perspective. • Understand global and local policies related to sustainable transportation, such as emissions standards and urban mobility plans.		

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	• Evaluate transportation projects using tools like environmental impact assessments and cost-benefit analysis. • Design and propose sustainable transportation systems, integrating multimodal approaches and clean energy solutions. • Use data analytics and modelling software to optimize traffic flow, reduce congestion, and improve fuel efficiency. • Exhibit a commitment to promoting sustainable and equitable transportation solutions. • Develop collaborative skills to work with policymakers, urban planners, and engineers in implementing transportation projects. • Foster a problem-solving mindset, focusing on innovative and practical approaches to reducing the environmental footprint of transportation.
Knowledge	• Principles of sustainable urban mobility, emissions reduction strategies, and energy efficiency in transportation. • Understanding of transport modes (public transit, non-motorized transport, electric vehicles) and their environmental implications. • Awareness of sustainable infrastructure, such as bike-sharing systems, pedestrian-friendly urban design, and electric charging networks. • Familiarity with environmental regulations and transportation standards, including those from international bodies like the IPCC and national governments. • Knowledge of emerging technologies like vehicle electrification, shared mobility platforms, and autonomous transport systems.
Skills	• Ability to perform assessments of transportation systems, identifying inefficiencies and proposing sustainable improvements. • Skills in developing long-term transportation plans that consider environmental, social, and economic factors. • Competence in designing low-carbon transportation solutions, from urban transit networks to infrastructure for electric vehicles. • Use of software tools for simulation and modelling of transportation scenarios to optimize energy use and reduce emissions. • Skills in managing transportation projects with a focus on sustainability, including stakeholder engagement and policy advocacy. • Understanding of how to develop policy recommendations that support sustainable transportation goals.

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Transverse / soft competencies	• Ability to collaborate with multidisciplinary teams, including urban planners, engineers, policymakers, and community leaders. • Strong communication skills to articulate the benefits and challenges of sustainable transportation solutions to diverse audiences. • A sense of social responsibility to advocate for equitable and environmentally conscious transportation systems. • Openness to new ideas and flexibility in adapting to technological and societal shifts in transportation. • Proficient in addressing complex transportation challenges with innovative, data-driven, and practical solutions.
Contents	• Emissions of fuels combustion/greenhouse gas emissions • Modal shift in passenger and freight transport to environmentally friendly transport modes • Transportation Choice & Technology • Multi/Intermodal Transport • Innovative Transport Concepts • Policies and measures by transport sector

Code	Title	Number of theoretical contact hours	Number of practical contact hours	Number of individual learning hours	ECTSs and µECTSs
C.T.#09/1-2	Sustainable Transportation	28	56	126	7
Type: Compulsory	Emissions of fuels combustion/greenhouse gas emissions	2	4	8	0,5
Domain: Transport	Modal shift in passenger and freight transport to environmentally friendly transport modes	4	8	18	1,0
Discipline: #09	Transportation choice & technology	6	12	28	1,5
Year: 1	Multi/Intermodal transport	6	12	28	1,5
Semester: 2	Innovative transport concepts	8	16	36	2,0
	Policies and measures by transport sector	2	4	8	0,5

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Hours	210	28	56	126	7
ECTSs	7				7,0

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3.2.3 Warehousing and Distribution Centres

Warehousing and Distribution Centres			ECTS: 5
Characteristics of the discipline	O.T.#15.1/2-3		
	Discipline type Compulsory /optional Optional	Discipline Pre-requisite No pre-requisites	Assessment summary Minimum Mark: 9.5 / 20
Full details of assessment and respective weight	Exam: 2h, + project, PBL, etc.		
Learning Method(s) and Delivery Mode(s)	According to D.2.5 and D.2.4		
Main Aim	The aim of the "Warehousing and Distribution Centres" discipline is to provide students with in-depth knowledge and practical skills to design, operate, and manage warehousing and distribution centres efficiently. The course focuses on optimizing storage, inventory, and material handling processes, with an emphasis on technologies, sustainability, and strategic decision-making to improve supply chain performance.		
Objectives	• To provide a thorough understanding of the role and functions of warehousing and distribution centres in modern supply chains. • To explore methods for improving space utilization, inventory management, and material handling within warehouse facilities. • To study the impact of emerging technologies (e.g., automation, robotics, and WMS software) in enhancing operational efficiency and accuracy. • To examine sustainable practices in warehousing, focusing on energy efficiency, waste reduction, and green logistics. • To understand and implement safety standards, risk management strategies, and compliance requirements in warehouse environments.		
Learning Outcomes	• Explain the core principles of warehousing, including layout design, storage methods, and distribution strategies. • Identify and describe key technologies and systems used in modern warehousing operations. • Understand environmental and regulatory considerations affecting warehousing and distribution practices. • Analyse and optimize warehouse layouts and material flow to improve efficiency and reduce costs. • Develop and implement inventory control systems and handling processes that align with operational goals.		

	• Use data and software tools to monitor performance and drive continuous improvement in warehousing operations. • Demonstrate a proactive approach to improving safety, sustainability, and efficiency in warehouse environments. • Collaborate effectively with team members to solve complex warehousing challenges. • Exhibit critical thinking and problem-solving abilities in managing the day-to-day and strategic aspects of warehousing.
Knowledge	• Understanding of warehousing types, design principles, and key functions, including storage, sorting, and order fulfilment. • Knowledge of distribution centre roles and their integration with transportation, logistics, and supply chain networks. • Familiarity with Warehouse Management Systems (WMS), automated storage and retrieval systems (AS/RS), and material handling equipment. • Awareness of trends like IoT, robotics, and data analytics in improving warehousing accuracy and productivity. • Knowledge of sustainable practices, energy-efficient operations, and waste reduction initiatives in warehousing. • Understanding of health, safety, and compliance requirements, including OSHA standards and environmental regulations.
Skills	• Ability to assess and optimize warehousing processes, including space utilization, picking strategies, and inventory control. • Proficiency in data-driven decision-making to enhance efficiency and reduce operational costs. • Competence in selecting and using WMS and other warehousing technologies for tracking, automation, and real-time monitoring. • Skills in implementing digital tools to increase accuracy and streamline order processing. • Ability to manage warehousing projects, ensuring effective planning, resource allocation, and adherence to safety standards. • Skills in risk assessment and mitigation to handle disruptions, safety concerns, and compliance issues.
Transverse / soft competencies	• Ability to work effectively within cross-functional teams, coordinating with logistics, transportation, and supply chain management professionals.

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	• Skills in taking a leadership role in warehouse operations, demonstrating initiative in problem-solving and continuous improvement efforts. • A commitment to ethical practices and sustainability in warehousing, promoting responsible resource use and minimal environmental impact. • Flexibility to adapt to evolving technologies, market demands, and operational challenges. • Proficiency in identifying challenges in warehouse operations and developing creative, practical solutions to improve processes.
Contents	• Energy Consumption and Efficiency in Warehouses • Sustainable Waste Management Practices • Green Building Practices and Certifications • Automation and IoT for Sustainable Warehousing • Eco-friendly Materials and Packaging Solutions

Code	Title	Number of theoretical contact hours	Number of practical contact hours	Number of individual learning hours	ECTSs and µECTSs
O.T.#15.1/2-3	Warehousing and Distribution Centres	14	28	108	5
Type: Optional	Energy consumption and efficiency in warehouses	2	4	16	0,7
Domain: Transport	Sustainable Waste Management practices	3	6	23	1,1
Discipline: #15	Green building practices and certifications	3	6	23	1,1
Year: 2	Automation and IoT for Sustainable Warehousing	3	6	23	1,1
Semester: 3	Eco-friendly materials and packaging solutions	3	6	23	1,1
Hours	150	14	28	108	5
ECTSs	5				5,0

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3.2.4 Greening Warehouses

Greening Warehouses			ECTS: 5
Characteristics of the discipline	O.T.#15.2/2-3		
	Discipline type Compulsory /optional Optional	Discipline Pre-requisite No pre-requisites	Assessment summary Minimum Mark: 9.5 / 20
Full details of assessment and respective weight	Exam: 2h, + project, PBL, etc.		
Learning Method(s) and Delivery Mode(s)	According to D.2.5 and D.2.4		
Main Aim	The aim of the "Greening Warehouses" discipline is to equip students with the knowledge and skills necessary to implement sustainable practices in warehouse operations. The course focuses on integrating environmental considerations into warehouse design, operations, and management, with an emphasis on reducing carbon footprints, enhancing energy efficiency, and promoting sustainability throughout the supply chain.		
Objectives	• To explore the principles of sustainability and their relevance to warehouse operations and supply chain management. • To analyse the environmental impacts of warehouse activities, including energy consumption, waste generation, and resource use. • To evaluate and adopt technologies that enhance energy efficiency and reduce emissions within warehouse facilities. • To create and implement operational strategies that promote sustainable practices, including waste reduction, recycling, and green procurement. • To understand relevant regulations and standards related to sustainability in warehousing, including LEED certification and other green building standards.		
Learning Outcomes	• Explain the key concepts of sustainability and their application to warehousing and logistics. • Identify the environmental impacts associated with warehouse operations and the importance of mitigating these impacts. • Understand the regulatory landscape and standards related to green warehousing and sustainable logistics. • Assess the energy and resource efficiency of warehouse operations and propose improvements.		

	• Develop and implement sustainable practices in warehouse design and operations, focusing on energy management and waste reduction. • Utilize tools and technologies for monitoring and reporting sustainability metrics in warehousing. • Demonstrate a commitment to sustainability and environmental responsibility in warehousing practices. • Collaborate with stakeholders to promote green initiatives and enhance sustainability in supply chains management. • Exhibit critical thinking in identifying and addressing sustainability challenges in warehouse operations.
Knowledge	• Understanding the principles of green logistics, sustainable supply chain management, and their application in warehousing. • Knowledge of life cycle assessment (LCA) and its relevance to warehouse operations and design. • Familiarity with local and global regulations regarding waste management, emissions, and sustainability in warehousing. • Awareness of certification programs such as LEED, BREEAM, and their criteria for green warehouses. • Understanding of energy-efficient technologies and practices, including renewable energy sources, smart lighting, and HVAC systems. • Knowledge of sustainable materials and their use in warehouse design and renovation.
Skills	• Ability to conduct energy audits and resource assessments to identify areas for improvement in warehouse sustainability. • Skills in developing and implementing energy management plans that optimize resource use and reduce waste. • Competence in managing projects aimed at greening warehouse operations, including planning, execution, and performance monitoring. • Skills in engaging stakeholders and communicating sustainability goals effectively. • Proficiency in using software tools to analyse sustainability data and track performance metrics related to energy consumption and waste reduction. • Ability to prepare reports and presentations that communicate sustainability efforts and outcomes to stakeholders.
Transverse / soft competencies	• Ability to work collaboratively with diverse teams, fostering a culture of sustainability in warehouse operations.

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	• Skills in leading initiatives to implement green practices in warehousing and encouraging a sustainable mindset among team members. • A strong commitment to ethical practices and social responsibility in promoting sustainable warehouse operations. • Openness to new ideas and technologies that enhance sustainability in warehousing and logistics. • Ability to critically analyse problems related to sustainability in warehousing and develop innovative solutions.
Contents	• Introduction to Sustainable Warehousing (an overview of sustainability in supply chain management, and the importance of greening warehouses in global logistics) • Sustainability principles and frameworks (core concepts of sustainability: environmental, social, and economic dimensions, and the frameworks and standards, e.g., LEED, ISO 14001, BREEAM) • Environmental impact assessment (methods for assessing the environmental impact of warehousing operations, Life Cycle Assessment (LCA) and its application in warehousing) • Sustainable warehouse design (principles of sustainable building design and retrofitting, and the use of eco-friendly materials and construction practices) • Technological innovations in green warehousing (overview of technologies supporting sustainable practices as automation, IoT, smart sensors, and the role of warehouse management systems (WMS) in sustainability) • Case studies of successful green warehousing

Code	Title	Number of theoretical contact hours	Number of practical contact hours	Number of individual learning hours	ECTSs and µECTSs
O.T.#15.2/2-3	Greening Warehouses	14	28	108	5
Type: Optional	Introduction to Sustainable Warehousing	2	4	16	0,7
Domain: Transport	Sustainability principles and frameworks	2	4	16	0,7
Discipline: #15	Environmental impact assessment	2	4	15	0,7
Year: 2	Sustainable warehouse design	2	4	15	0,7
Semester: 3	Technological innovations in green warehousing	3	6	23	1,1

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	Case studies of successful green warehousing	3	6	23	1,1
Hours	150	14	28	108	5
ECTSs	5				5,0

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3.2.5 Optimization in Warehousing

Optimization in Warehousing			ECTS: 5
Characteristics of the discipline	O.T.#15.3/2-3		
	Discipline type Compulsory /optional Optional	Discipline Pre-requisite No pre-requisites	Assessment summary Minimum Mark: 9.5 / 20
Full details of assessment and respective weight	Exam: 2h, + project, PBL, etc.		
Learning Method(s) and Delivery Mode(s)	According to D.2.5 and D.2.4		
Main Aim	The aim of the "Optimization in Warehousing" discipline is to equip students with the analytical skills and knowledge necessary to enhance the efficiency and effectiveness of warehousing operations. The course focuses on applying optimization techniques and technologies to improve inventory management, space utilization, and overall warehouse performance in a dynamic supply chain environment.		
Objectives	• To provide a thorough understanding of optimization principles and their application in warehouse management. • To assess and design warehouse layouts that maximize space utilization and operational efficiency. • To explore methods for effective inventory control, including demand forecasting, reorder points, and safety stock levels. • To evaluate the role of advanced technologies (e.g., automation, data analytics, and artificial intelligence) in optimizing warehouse operations. • To establish key performance indicators (KPIs) and continuous improvement strategies to enhance warehouse productivity.		
Learning Outcomes	• Explain the fundamental concepts of warehouse optimization and its impact on supply chain efficiency. • Identify various optimization techniques and tools applicable to warehousing operations. • Demonstrate a commitment to continuous improvement and data-driven decision-making in warehouse operations. • Collaborate effectively with cross-functional teams to implement optimization initiatives. • Exhibit critical thinking and problem-solving skills in addressing operational challenges in warehousing.		

Knowledge	• Understanding of optimization principles, including linear programming, simulation, and heuristic methods. • Knowledge of the significance of optimizing workflows, inventory levels, and resource allocation in warehousing. • Familiarity with various warehouse layout types (e.g., U-shape, straight-line, and grid) and their impact on operations. • Awareness of flow analysis and space utilization techniques to improve efficiency. • Understanding of various inventory management models (e.g., ABC analysis, JIT, and VMI) and their application in warehousing.
Skills	• Ability to perform workflow analysis and redesign warehouse layouts for optimal space utilization. • Skills in conducting time and motion studies to identify inefficiencies and areas for improvement. • Proficiency in using data analytics tools to analyse historical data, forecast demand, and optimize inventory levels. • Skills in employing software for simulation and modelling of warehouse operations. • Ability to implement and leverage technology solutions (e.g., WMS, automation) to enhance warehouse efficiency. • Skills in integrating IoT devices and sensors for real-time monitoring and performance optimization.
Transverse / soft competencies	• Ability to work collaboratively with logistics, operations, and IT teams to drive optimization initiatives • Skills in leading optimization projects and fostering a culture of continuous improvement within the warehouse. • Proficient in identifying problems, analysing data, and developing innovative solutions to optimize warehouse operations. • Openness to adopting new technologies and methodologies that enhance warehousing efficiency. • Strong communication skills to articulate optimization strategies and results to stakeholders at all levels.
Contents	• Introduction to Warehouse Optimization • Warehouse Layout and Design Optimization • Inventory Management Optimization • Data Analytics in Warehousing • Technology and Automation in Warehousing • Performance Metrics and KPIs • Continuous Improvement Strategies

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	• Case Studies in Warehouse Optimization

Code	Title	Number of theoretical contact hours	Number of practical contact hours	Number of individual learning hours	ECTSs and µECTSs
O.T.#15.3/2-3	Optimization in Warehousing	14	28	108	5
Type: Optional	Introduction to Warehouse Optimization	2	4	16	0,7
Domain: Transport	Warehouse Layout and Design Optimization	2	4	16	0,7
Discipline: #15	Inventory Management Optimization	2	4	16	0,7
Year: 2	Data Analytics in Warehousing	1	2	8	0,4
Semester: 3	Technology and Automation in Warehousing	2	4	15	0,7
	Performance Metrics and KPIs	1	2	7	0,3
	Continuous Improvement Strategies	1	2	7	0,3
	Case Studies in Warehouse Optimization	3	6	23	1,1
Hours	150	14	28	108	5
ECTSs	5				5,0

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3.3 Forms for Energy production disciplines

3.3.1 Sustainable energy systems and environmental impact

Sustainable energy systems and environmental impact			ECTS: 7
Characteristics of the discipline	C.E.#10/1-2		
	Discipline type Compulsory /optional	Discipline Pre-requisite	Assessment summary
	Compulsory	No pre-requisites	Minimum Mark: 9.5 / 20
Full details of assessment and respective weight	Exam: 2h, + project, PBL, etc.		
Learning Method(s) and Delivery Mode(s)	According to D.2.5 and D.2.4		
Main Aim	The aim of the "Sustainable Energy Systems and Environmental Impact" discipline is to provide students with a comprehensive understanding of sustainable energy technologies and their environmental implications. The course focuses on the integration of renewable energy sources, energy efficiency measures, and environmental assessment methods to promote sustainable energy practices that minimize ecological footprints.		
Objectives	• To explore the principles of sustainable energy systems, including renewable energy sources and energy efficiency strategies. • To evaluate the environmental impacts of various energy systems and understand methodologies for conducting environmental assessments. • To assess the role and integration of renewable energy technologies, such as solar, wind, hydro, and biomass, in sustainable energy systems. • To create and implement strategies that enhance energy efficiency in various sectors, including residential, industrial, and transportation. • To analyse the policy and regulatory frameworks that support sustainable energy development and address environmental impacts.		
Learning Outcomes	• Explain key concepts and principles of sustainable energy systems and their importance in reducing environmental impacts.		

	• Identify and describe various renewable energy technologies and their applications in different contexts. • Assess the environmental impacts associated with energy systems using appropriate methodologies and tools. • Develop and propose energy efficiency strategies that can be implemented in various sectors. • Demonstrate a commitment to sustainability and environmental stewardship in energy practices. • Collaborate effectively with stakeholders to promote sustainable energy solutions and practices.
Knowledge	• In-depth knowledge of various energy production systems including their mechanisms, efficiencies, and environmental impacts • Knowledge of techniques and technologies to mitigate pollution from energy production • Proficiency in analysing greenhouse gas emissions from various energy production systems • Understanding of the strategies to mitigate the impact of energy production on climate change • In-depth knowledge of various renewable energy technologies and their operational principles • Understanding of the development and implementation of sustainable energy policies and regulations • Knowledge of practical applications and successful examples of sustainable energy projects • Understanding of the role and challenges of transitioning from traditional to renewable energy sources • Understanding of the broader implications of pollution on environmental health and ecosystems
Skills	• Ability to compare and contrast the environmental, economic, and social impacts of different energy systems • Ability to critically evaluate the sustainability of energy systems using LCA methodologies • Competence in using LCA results to inform sustainable decision-making in energy production • Ability to innovate and improve the efficiency and sustainability of renewable energy technologies • Ability to identify and implement best practices in sustainable energy development • Skills in managing sustainable energy projects from conception to completion • Skills in conducting life cycle assessments (LCA) to evaluate the environmental impacts of different energy systems from cradle to grave • Skills in assessing the impacts of energy production on air, water, and soil quality • Competence in planning and implementing renewable energy projects

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	• Ability to advocate for sustainable energy policies and effectively communicate their importance • Skills in ensuring compliance with national and international energy regulations • Ability to relate GHG emissions to international climate policies and agreements • Skills in managing and evaluating sustainable energy projects effectively.
Transverse / soft competencies	• Comprehensive understanding of the interdisciplinary nature of energy systems and their environmental impacts. • Proficiency in analysing complex data related to energy production and environmental impacts. • Ability to identify, analyse, and solve problems related to unsustainable energy practices. • Competence in developing innovative solutions to enhance the sustainability of energy systems. • Technical knowledge of renewable energy technologies and sustainable energy practices. • Understanding of the regulatory landscape and ability to develop policies promoting sustainable energy. • Effective communication skills to advocate for and educate about sustainable energy systems and their environmental benefits. • Leadership abilities to drive and manage environmental risk assessment and mitigation efforts. • Leadership abilities to drive and manage transitions towards more sustainable energy systems.
Contents	• Overview of energy production systems: fossil fuels, nuclear, and renewables • Life cycle analysis of energy systems • Greenhouse gas emissions and climate change • Air, water, and soil pollution from energy production • Renewable energy technologies (solar, wind, hydro, and biomass) • Sustainable energy policies and regulations • Case studies on successful sustainable energy projects

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Code	Title	Number of theoretical contact hours	Number of practical contact hours	Number of individual learning hours	ECTSs and µECTSs
C.E.#10/1-2	Sustainable Energy Systems and Environmental Impact	28	56	126	7
Type: Compulsory	Overview of energy production systems: fossil fuels, nuclear, and renewables	4	8	18	1
Domain: Energy	Life cycle analysis of energy systems	4	8	18	1
Discipline: #10	Greenhouse gas emissions and climate change	4	8	18	1
Year: 1	Air, water, and soil pollution from energy production	4	8	18	1
Semester: 2	Renewable energy technologies (solar, wind, hydro, and biomass)	4	8	18	1
	Sustainable energy policies and regulations	4	8	18	1
	Case studies on successful sustainable energy projects	4	8	18	1
Hours	210	28	56	126	7
ECTSs	7				7,0

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3.3.2 Climate change mitigation and adaptation in the energy

Climate change mitigation and adaptation in the energy sector			ECTS: 5
Characteristics of the discipline	C.E.#16/2-3		
	Discipline type Compulsory /optional	Discipline Pre-requisite	Assessment summary
	Compulsory	No pre-requisites	Minimum Mark: 9.5 / 20
Full details of assessment and respective weight	Exam: 2h, + project, PBL, etc.		
Learning Method(s) and Delivery Mode(s)	According to D.2.5 and D.2.4		
Main Aim	The aim of the "Climate change mitigation and adaptation in the energy sector" discipline is to equip students with the knowledge and skills necessary to understand and address the challenges posed by climate change within the energy sector. The course focuses on strategies for mitigating greenhouse gas emissions and adapting energy systems to enhance resilience against climate change impacts.		
Objectives	• To provide a foundational understanding of climate change science, its causes, and its impacts on energy systems and the environment. • To analyse various mitigation strategies aimed at reducing greenhouse gas emissions from energy production and consumption. • To assess adaptation measures that can enhance the resilience of energy systems to the effects of climate change, including extreme weather events and changing climate conditions. • To examine the role of international agreements, national policies, and local regulations in promoting climate change mitigation and adaptation in the energy sector. • To develop skills for engaging with stakeholders and facilitating collaboration among diverse groups to implement climate action strategies in the energy sector.		
Learning Outcomes	• Explain the science of climate change, including its causes, consequences, and relevance to the energy sector. • Identify and describe key mitigation and adaptation strategies specific to energy systems. • Assess the effectiveness of various mitigation measures and adaptation strategies in reducing vulnerability and enhancing resilience in the energy sector. • Develop action plans that incorporate mitigation and adaptation strategies tailored to specific energy contexts.		

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	• Demonstrate a proactive attitude towards climate change challenges and a commitment to sustainable energy practices. • Collaborate with stakeholders from various sectors to promote climate change mitigation and adaptation initiatives.
Knowledge	• Comprehensive understanding of climate science fundamentals and how the energy sector contributes to climate change • Proficiency in creating and managing greenhouse gas inventories • Understanding of international climate agreements (e.g., Paris Agreement) and their implications for the energy sector • In-depth knowledge of various renewable energy technologies (solar, wind, hydro, biomass) • Knowledge of energy efficiency measures and technologies applicable in various energy systems • Knowledge of strategies to adapt energy infrastructure to climate change impacts • Data analysis for identifying and assessing environmental risks associated with energy production. • Knowledge of environmental regulations and international standards related to environmental risk management in the energy sector (e.g., ISO 14001, EU regulations).
Skills	• Skills in analysing the energy sector's role within the broader context of climate systems • Ability to measure and report greenhouse gas emissions accurately • Skills in ensuring energy projects comply with national and international climate policies • Skills in integrating renewable energy sources into existing grid systems • Ability to design and implement energy efficiency projects • Ability to plan and manage CCUS projects • Ability to enhance the resilience of energy systems against climate-related disruptions • Ability to conduct environmental risk assessments using recognized methodologies (such as Life Cycle Assessment or Ecological Risk Assessment). • Ability to assess the impact of different energy production methods on climate change • Skills in conducting cost-benefit analyses for energy efficiency investments • Skills in evaluating and implementing CCUS solutions in different energy production contexts • Skills in assessing and mitigating climate risks to energy infrastructure

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Transverse / soft competencies	• Proficiency in understanding and applying climate science principles to the energy sector. • Competence in managing greenhouse gas inventories and accounting. • Skills in navigating and implementing international and national climate policies. • Expertise in integrating renewable energy sources into grid systems and managing related challenges. • Knowledge and skills in designing and implementing effective energy efficiency measures. • Problem-solving applied to risk management, to identify innovative solutions that reduce environmental impact without compromising energy production. • Technical and managerial skills in deploying carbon capture, utilization, and storage technologies. • Ability to develop and implement strategies to adapt energy infrastructure to climate change impacts. • Understanding of and ability to ensure compliance with relevant climate and energy regulations. • Leadership skills in promoting and managing sustainable energy projects. • Strong analytical and critical thinking skills to evaluate and address climate change challenges in the energy sector. • Skills in managing complex projects related to climate change mitigation and adaptation. • Ability to effectively engage with stakeholders in the development and implementation of climate strategies.
Contents	• Climate science and the role of the energy sector • Greenhouse gas inventories and accounting • International climate agreements and national policies • Renewable energy integration and grid management • Energy efficiency measures and technologies • Carbon capture, utilization, and storage (CCUS) • Climate adaptation strategies for the energy infrastructure • Environmental Risk Assessment and Management in Energy Production

Code	Title	Number of theoretical contact hours	Number of practical contact hours	Number of individual learning hours	ECTSs and µECTSs

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C.E.#16/2-3	Climate Change Mitigation and Adaptation in the Energy Sector	14	28	108	5
Type: Compulsory	Climate science and the role of the energy sector	2	4	12	0,56
Domain: Energy	Greenhouse gas inventories and accounting	2	4	15	0,69
Discipline: #16	International climate agreements and national policies	1,5	3	13	0,6
Year: 2	Renewable energy integration and grid management	2	4	15	0,69
Semester: 3	Energy efficiency measures and technologies	1,5	3	15	0,69
	Carbon capture, utilization, and storage (CCUS)	2	4	11	0,51
	Climate adaptation strategies for the energy infrastructure	1,5	3	13	0,6
	Environmental risk assessment and management in energy production	1,5	3	14	0,65
Hours	150	14	28	108	5
ECTSs	5				5,0

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3.4 Forms for Waste disciplines

3.4.1 Waste Management

Waste Management			ECTS: 5
Characteristics of the discipline	C.W.#04/1-1		
	Discipline type Compulsory /optional	Discipline Pre-requisite	Assessment summary
	Compulsory	No pre-requisites	Minimum Mark: 9.5 / 20
Full details of assessment and respective weight	Exam: 2h, + project, PBL, etc.		
Learning Method(s) and Delivery Mode(s)	According to D.2.5 and D.2.4		
Main Aim	The aim of the "Waste Management" discipline is to provide students with a comprehensive understanding of waste management practices and strategies. The course focuses on the principles, policies, and practical methods for effectively managing waste to minimize environmental impact and promote sustainability.		
Objectives	• To explore the concepts and principles underlying effective waste management systems. • To analyse current waste management practices, including waste reduction, recycling, and safe disposal. • To understand international, national, and local policies governing waste management and their implementation. • To develop strategies for promoting sustainable waste management in various sectors. • To evaluate the environmental impact of different waste streams and propose solutions to mitigate these effects.		
Learning Outcomes	• Explain key concepts, strategies, and frameworks for effective waste management. • Identify various types of waste and their impact on the environment and human health. • Design and implement waste management plans that reduce environmental impact. • Assess and recommend improvements to the waste management systems using sustainability principles. • Demonstrate an ethical commitment to environmental responsibility and sustainability in waste management practices. • Collaborate effectively with community stakeholders to promote waste reduction and management strategies.		

Knowledge	• Acquire a thorough understanding of different waste classification systems and their application in waste management practices. • Analyse and propose technologies that can be applied to recycling and recovering waste, in order to obtain the maximum potential resource from it, also taking into account the environmental and social component. • Understand the principles and practices of municipal waste management, including collection, transportation, treatment, and disposal. • Select the technologies that can be applied for the decontamination and regeneration of urban waste and determine the possibilities of reusing treated water, solids and gases generated. • Gain comprehensive knowledge of industrial solid waste management practices, including waste minimization, recycling, and regulatory compliance. • Understand the regulatory framework governing hazardous waste management, including classification, handling, treatment, and disposal. • Use tools to predict and validate waste behaviour in each of the indicated disposal or dumping systems. • Acquire detailed knowledge of various wastewater treatment technologies and their applications in different contexts. • Acquire the use of laboratory techniques for monitoring and controlling waste and wastewater, taking into account the quality criteria established by current legislation. • Understand the European waste management regulations, including waste obligations, compliance requirements, and the impact of these regulations on waste management practices. • Fundamentals of energy, energy use in an industrial society. Fossil fuels. Renewable energy: solar, wind, hydro, biomass, ocean energy, geothermal, hydrogen technologies. Nuclear energy: status and problems. Energy planning. Energy conservation. Energy transport. Local and global effects of energy use. • Know the legislation and be able to apply specific regulations at different levels: community, state, autonomous and local. • Have a knowledge base of fundamental concepts that apply to waste management. • Differentiate between different types and fractions, as well as treatment flows based on the management and treatment hierarchy. • Analyse legislation and its application at different levels: community, state, autonomous and local. • Evaluate planning aspects and waste management strategies at different levels. • Understand the social, economic and environmental responsibility involved in waste management.

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Skills	• Use advanced techniques to accurately classify different types of waste, ensuring appropriate processing and disposal methods are applied. • Assess and establish the application of available technologies for recycling and recovering waste. • Implement innovative strategies to reduce municipal waste generation, promote recycling, and increase community engagement in sustainable practices. • Explain the different treatments and technologies applicable to the purification of urban wastewater and its possible reuse, including the use of the solid waste generated. • Apply advanced treatment methods to manage industrial solid waste, including recycling, composting, and safe disposal practices. • Execute effective treatment and disposal methods for hazardous waste, employing best practices to mitigate environmental and health risks. • Interpret and use waste coding and characterization to select appropriate management taking into account safety and protection criteria. • Operate and optimize wastewater treatment plants, using state-of-the-art technologies and processes to ensure efficient and effective wastewater management. • Promote interaction with professionals and researchers and learn about specific cases of wastewater treatment and reuse. Implement and manage technologies for energy recovery from waste, such as anaerobic digestion and incineration, to enhance resource efficiency. • To understand the basic concepts, sources, applications and problems of interest related to the use of energy and its interaction with the environment. • Renewable energies are given special attention. • Study singular cases of management, conventional and non-conventional, considering waste as resources and analysing management strategies in each case. • Analyse legislation and its application at different levels: community, state, autonomous and local. Evaluate planning aspects and waste management strategies at different levels. Understand the social, economic and environmental responsibility involved in waste management. • Develop and oversee comprehensive municipal waste management programs, ensuring efficient collection, processing, and disposal while minimizing environmental impact. • Acquire the use of laboratory techniques for monitoring and controlling waste and wastewater, taking into account the quality criteria established by current legislation.

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	• Design and implement sustainable industrial solid waste management systems that align with industry best practices and regulatory requirements. • Lead initiatives for the safe handling, transportation, and disposal of hazardous waste, ensuring compliance with health, safety, and environmental regulations. • Evaluate prior treatments necessary to facilitate the storage, transformation or disposal of waste in an environmentally safe and socially acceptable manner. • Manage projects that convert waste to energy, optimizing the use of available technologies to enhance energy recovery and reduce landfill dependence. • Provide analytical tools and methodologies for wastewater quality control. • Ensure that all waste management activities within an organization comply with local, national, and European regulations, including accurate reporting and documentation. • Planning capacity to diagnose, evaluate and optimize the production and use of energy taking into account available resources, the “principle of energy solidarity” and the needs of developing societies.
Transverse / soft competencies	• Demonstrate the ability to communicate complex waste management concepts and regulations clearly and effectively to diverse stakeholders, including policymakers, industry professionals, and the general public. • Be able to assess complex situations and propose solutions in the field of waste management, wastewater and its potential recycling or regeneration, taking into account the possible social, economic and environmental repercussions. • Apply critical thinking and problem-solving skills to identify, analyse, and resolve challenges in waste classification, municipal and industrial waste management, hazardous waste handling, wastewater treatment, and waste-to-energy conversion. • Know how to interpret the different parameters of waste characterization in accordance with standardized methodologies, as well as to know how to differentiate and apply the most appropriate analytical techniques in order to understand their properties for their subsequent treatment and management. • Exhibit strong leadership and team collaboration skills, guiding multidisciplinary teams to achieve common goals in waste management projects and ensuring effective teamwork and resource allocation. • Know and apply local, regional, national and international regulations and rules in the general field of protection of human health and the environment.

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- Make ethical and sustainable decisions in waste management practices, considering the long-term environmental, social, and economic impacts of waste treatment and disposal methods.
- Design experiments that allow simulating environmental protection systems installed for the safe confinement of waste that is dangerous in terms of being potential contaminants.
- Manage complex waste management projects efficiently, utilizing advanced project management techniques to plan, execute, monitor, and evaluate projects within set timelines and budgets.
- Acquire the use of laboratory techniques for monitoring and controlling waste and wastewater, taking into account the quality criteria established by current legislation.
- Show adaptability to evolving waste management technologies and regulations and commit to continuous professional development and lifelong learning to stay updated with industry advancements and best practices.
- Be able to integrate knowledge and face the complexity of formulating judgments based on information that, although incomplete or limited, includes reflections on the social and ethical responsibilities linked to the application of their knowledge and judgments.
- Be able to develop their professional activity while respecting fundamental rights and equal opportunities, within the framework of professional ethics and ethical commitment.
- Know and apply local, regional, national and international regulations and rules in the general field of protection of human health and the environment.
- Demonstrate the ability to communicate complex waste management concepts and regulations clearly and effectively to diverse stakeholders, including policymakers, industry professionals, and the general public.
- Be able to assess complex situations and propose solutions in the field of waste management, wastewater and its potential recycling or regeneration, taking into account the possible social, economic and environmental repercussions.
- Apply critical thinking and problem-solving skills to identify, analyse, and resolve challenges in waste classification, municipal and industrial waste management, hazardous waste handling, wastewater treatment, and waste-to-energy conversion.
- To know how to interpret the different parameters of waste characterization in accordance with standardized methodologies, as well as to know how to differentiate and apply the most appropriate analytical techniques in order to understand their properties for their subsequent treatment and management.
- Exhibit strong leadership and team collaboration skills, guiding multidisciplinary teams to achieve common goals in waste management projects and ensuring effective teamwork and resource allocation. Know and

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	apply local, regional, national and international regulations and rules in the general field of protection of human health and the environment. • Make ethical and sustainable decisions in waste management practices, considering the long-term environmental, social, and economic impacts of waste treatment and disposal methods. • Design experiments that allow simulating environmental protection systems installed for the safe confinement of waste that is dangerous in terms of being potential contaminants. • Manage complex waste management projects efficiently, utilizing advanced project management techniques to plan, execute, monitor, and evaluate projects within set timelines and budgets. • Acquire the use of laboratory techniques for monitoring and controlling waste and wastewater, taking into account the quality criteria established by current legislation. • Show adaptability to evolving waste management technologies and regulations, and commit to continuous professional development and lifelong learning to stay updated with industry advancements and best practices. • Be able to integrate knowledge and face the complexity of formulating judgments based on information that, although incomplete or limited, includes reflections on the social and ethical responsibilities linked to the application of their knowledge and judgments. • Be able to develop their professional activity while respecting fundamental rights and equal opportunities, within the framework of professional ethics and ethical commitment. • Know and apply local, regional, national and international regulations and rules in the general field of protection of human health and the environment.
Contents	• Waste Classification • Municipal waste • Industrial solid waste • Hazard waste • Waste water • Waste to energy • Waste regulations and compliance

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Code	Title	Number of theoretical contact hours	Number of practical contact hours	Number of individual learning hours	ECTSs and µECTSs
C.W.#04/1-1	Waste Management	14	28	108	5
Type: Compulsory	Waste classification	2	4	15	0,7
Domain: Waste management	Municipal waste	2	4	16	0,7
Discipline: #04	Industrial solid waste	2	4	16	0,7
Year: 1	Hazard waste	2	4	16	0,7
Semester: 1	Waste water	2	4	15	0,7
	Waste to energy	2	4	15	0,7
	Waste regulations and compliance	2	4	15	0,7
Hours	150	14	28	108	5
ECTSs	5				5,0

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3.4.2 Waste Technologies

Waste technologies			ECTS: 7
Characteristics of the discipline	C.W.#08/1-2		
	Discipline type Compulsory /optional	Discipline Pre-requisite	Assessment summary
	Compulsory	No pre-requisites	Minimum Mark: 9.5 / 20
Full details of assessment and respective weight	Exam: 2h, + project, PBL, etc.		
Learning Method(s) and Delivery Mode(s)	According to D.2.5 and D.2.4		
Main Aim	The aim of the "Waste Technologies" discipline is to familiarize students with advanced technologies used for waste treatment, recycling, and energy recovery. The course emphasizes the role of innovative technologies in transforming waste into resources and reducing the overall environmental impact.		
Objectives	• To understand the role of technology in improving waste treatment and resource recovery. • To study various waste processing technologies, including mechanical, biological, and thermal methods. • To evaluate the effectiveness and environmental impact of different waste treatment technologies. • To integrate waste technologies into circular economy frameworks for sustainable resource management. • To learn how to select appropriate waste technologies based on waste type, regulatory requirements, and sustainability goals.		
Learning Outcomes	• Describe the main technologies used for waste treatment, recycling, and energy recovery. • Understand the environmental and economic impacts of adopting specific waste technologies. • Assess the suitability of different waste technologies for specific waste streams and environmental contexts. • Design technology-driven solutions for effective waste management and resource recovery. • Demonstrate an innovative and solution-oriented mindset toward waste technology applications. • Work collaboratively with engineers, environmental scientists, and policymakers to implement effective waste technology solutions.		

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Knowledge	• Acquire comprehensive knowledge of various waste technology systems and their applications in waste management • Understand the principles and technologies behind wastewater processing methods, including biological, chemical, and physical treatment processes. • Gain in-depth knowledge of engineering principles and practices in solid waste management, covering collection, transportation, treatment, and disposal. • Understand the technologies and methods used in air quality control, including filtration, scrubbing, and emissions monitoring. • Acquire detailed knowledge of energy technologies that address environmental challenges, focusing on renewable energy sources and waste-to-energy systems. • Understand the concepts and practices of sustainable production processes and circular business models, emphasizing resource efficiency and waste minimization. • To analyse and evaluate waste management from an environmental and socio-economic perspective using input-output and externality analysis methodologies. • To conceptualise and develop a life cycle analysis (LCA) in a waste treatment plant, considering the subsequent inventory (LCI) and its environmental behaviour and potential impact.
Skills	• Apply advanced waste technologies effectively to improve waste management practices and reduce environmental impact. • Study the technologies to be applied for the decontamination and regeneration of industrial and agricultural wastewater. • Operate and maintain wastewater treatment systems, ensuring optimal performance and adherence to environmental regulations. • Develop and execute engineering solutions for solid waste management, including recycling, composting, and safe disposal techniques. • Implement air quality control measures to monitor and reduce air pollutants from waste management activities. • Execute energy recovery projects from waste, utilizing technologies such as anaerobic digestion and incineration to produce renewable energy. • Apply circular economy principles in business models to create sustainable production and consumption systems, reducing waste and promoting resource reuse.

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	• Be able to propose improvements to reduce waste, avoiding its final disposal and establishing lines of work to achieve the end of the waste condition. • Demonstrate the ability to integrate various waste technologies into comprehensive waste management strategies that enhance efficiency and sustainability. • Manage and optimize wastewater processing technologies to ensure effective treatment and compliance with environmental standards. • Design and implement engineering solutions for solid waste management, balancing technical feasibility with environmental and economic considerations. • Apply advanced air quality control technologies to minimize emissions and pollutants from waste management processes. • Promote and implement sustainable production processes within industries, reducing waste generation and enhancing resource efficiency. • Develop waste management and treatment projects focusing on the application of life cycle analysis and circular economy concepts.
Transverse / soft competencies	• Demonstrate the ability to collaborate effectively with professionals from various disciplines, including engineering, environmental science, policy-making, and business, to develop and implement integrated waste technology solutions. • Exhibit strong communication skills to advocate for advanced waste technologies and sustainable practices, effectively conveying complex technical information to non-specialist stakeholders and the general public. • Apply critical thinking and innovative problem-solving skills to identify and develop creative solutions to challenges in waste management, including wastewater processing, solid waste engineering, air quality control, and energy recovery. • Lead and manage waste technology projects efficiently, demonstrating advanced project management skills to plan, execute, monitor, and evaluate projects within set timelines and budgets, while motivating and guiding teams towards achieving project goals. • Make informed and ethical decisions that prioritize sustainability and environmental responsibility in the application and development of waste technologies, balancing economic, social, and environmental considerations. • Show adaptability to rapidly changing technologies and regulatory landscapes in the field of waste management, and commit to continuous professional development and lifelong learning to stay current with industry advancements and best practices. Collect, compare, analyse and communicate information in the field

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	of waste management and resource recovery, using various sources including the application of information and communication technologies. • Demonstrate the ability to collaborate effectively with professionals from various disciplines, including engineering, environmental science, policy-making, and business, to develop and implement integrated waste technology solutions. • Exhibit strong communication skills to advocate for advanced waste technologies and sustainable practices, effectively conveying complex technical information to non-specialist stakeholders and the general public. • Show adaptability to rapidly changing technologies and regulatory landscapes in the field of waste management. • Collect, compare, analyse and communicate information in the field of waste management and resource recovery, using various sources including the application of information and communication technologies.
Contents	• Waste water processing technology • Solid waste engineering and management • Air quality control technologies • Sustainable production processes • Introduction to circular business models

Code	Title	Number of theoretical contact hours	Number of practical contact hours	Number of individual learning hours	ECTSs and µECTSs
C.W.#08/1-2	Waste Technologies	28	56	126	7
Type: Compulsory	Waste water processing technology	4	8	18	1,0
Domain: Waste management	Solid waste engineering and management	4	8	18	1,0
Discipline: #08	Air quality control technologies	4	8	18	1,0
Year: 1	Energy technologies for solving environmental challenges	4	8	18	1,0
Semester: 2	Sustainable production processes	8	16	36	2,0
	Introduction to circular business models	4	8	18	1,0
Hours	210	28	56	126	7
ECTSs	7				7,0

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3.5 Forms for Residential activities disciplines

3.5.1 Sustainable residential activities

Sustainable residential activities			ECTS: 6
Characteristics of the discipline	C.R.#14/2-3		
	Discipline type Compulsory /optional	Discipline Pre-requisite	Assessment summary
	Compulsory	No pre-requisites	Minimum Mark: 9.5 / 20
Full details of assessment and respective weight	Exam: 2h, + project, PBL, etc.		
Learning Method(s) and Delivery Mode(s)	According to D.2.5 and D.2.4		
Main Aim	The aim of the "Sustainable Residential Activities" discipline is to equip students with the knowledge and practical skills needed to develop and promote sustainable practices in residential settings. The course emphasizes strategies for improving energy efficiency, water conservation, waste management, and overall environmental impact in residential communities.		
Objectives	• To explore the core principles of sustainability and how they apply to residential living environments. • To study and implement strategies for improving energy and resource efficiency in residential buildings and activities. • To evaluate waste reduction and recycling methods suitable for residential communities and their environmental benefits. • To develop and promote strategies for efficient water use and management in residential areas. • To learn techniques for involving and educating residents about sustainable living and behaviour changes.		
Learning Outcomes	• Explain the importance of sustainability in residential settings and the environmental challenges related to residential activities. • Identify and describe energy-efficient technologies, waste management systems, and water-saving practices applicable to residential areas. • Design and implement sustainable practices for residential energy use, water conservation, and waste management.		

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	• Conduct assessments of residential activities to identify areas for sustainability improvements. • Demonstrate a proactive attitude towards environmental stewardship in residential contexts. • Collaborate effectively with homeowners, community groups, and local governments to promote sustainable living practices. • Show adaptability and willingness to learn and apply new sustainability strategies and technologies.
Knowledge	• Knowledge of renewable energy systems for residential use (e.g., solar panels, wind turbines). • Understanding of water conservation strategies and technologies for residential applications (e.g., low-flow fixtures, rainwater harvesting). • Awareness of sustainable food choices, including organic, locally sourced, and seasonal foods. • Knowledge of eco-friendly household products and alternatives to conventional cleaners and chemicals. • Knowledge of materials selection, insulation techniques, and passive design strategies to improve building performance • Knowledge of smart home technologies and automation for optimizing energy use and enhancing user comfort • Knowledge of water-efficient landscaping and irrigation practices to minimize water use in residential settings. • Understanding of green building principles and sustainable construction practices for residential buildings.
Skills	• Skills in implementing energy-efficient technologies and practices to reduce energy consumption and promote sustainability. • Ability to assess water consumption patterns and identify opportunities for water conservation. • Skills in promoting sustainable diets and lifestyles that prioritize environmental and health considerations. • Ability to assess the environmental impact of household products and materials (e.g., biodegradability, recyclability). • Skills in retrofitting existing homes for energy efficiency and sustainability. • Ability to recommend energy-efficient appliances and devices to reduce electricity usage. • Competence in assessing energy consumption patterns and conducting energy audits in residential buildings. • Competence in reducing food waste through composting and mindful consumption practices. • Competence in selecting and using sustainable products that contribute to indoor air quality and environmental sustainability

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	• Competence in managing energy consumption related to electronics and appliances in residential settings.
Transverse / soft competencies	• Analytical skills to evaluate residential sustainability practices and identify improvement opportunities. • Ability to address complex challenges related to energy efficiency, water conservation, and sustainable living. • Capacity to develop innovative solutions that balance environmental, economic, and social considerations • Effective communication of sustainable living practices and their benefits to household members and communities. • Ability to articulate the importance of sustainable residential activities and motivate behaviour change. • Skills in fostering dialogue and collaboration with family members, neighbours, and community groups on sustainability initiatives • Collaboration with household members and stakeholders (e.g., contractors, energy auditors) to implement sustainable practices. • Ability to coordinate efforts for household sustainability projects and initiatives. • Skills in building consensus and engaging diverse perspectives to achieve sustainability goals • Leadership in promoting sustainable behaviours and practices within residential settings. • Ability to influence household decisions and actions towards greater sustainability. • Capacity to lead by example and inspire others to adopt environmentally responsible behaviours • Adaptation to changing environmental conditions, technological advancements, and household needs. • Flexibility in adjusting lifestyle habits and practices to align with sustainable living principles. • Ability to explore and adopt new ideas and innovations for enhancing residential sustainability • Understanding of ethical considerations in residential sustainability, including resource conservation and environmental stewardship. • Commitment to promoting social equity, health, and well-being through sustainable residential activities. • Awareness of the broader impacts of residential choices on the environment, community, and future generations
Contents	• Domestic energy production and consumption • Domestic water usage • Food consumption • Household products • Building and renovation

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	• Electronic use

Code	Title	Number of theoretical contact hours	Number of practical contact hours	Number of individual learning hours	ECTSs and µECTSs
C.R.#14/2-3	Sustainable residential activities	28	28	124	6
Type: Compulsory	Domestic energy production and consumption	5	5	22	1,1
Domain: Residential	Domestic water usage	4	4	18	0,9
Discipline: #14	Food consumption	5	5	22	1,1
Year: 2	Household products	5	5	22	1,1
Semester: 3	Building and renovation	5	5	22	1,1
	Electronic use	4	4	18	0,9
Hours	180	28	28	124	6
ECTSs	6				6,0

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3.6 Forms for Agriculture disciplines

3.6.1 Sustainable

Sustainable agriculture			ECTS: 6
Characteristics of the discipline	C.R.#17/2-3		
	Discipline type Compulsory /optional	Discipline Pre-requisite	Assessment summary
	Compulsory	No pre-requisites	Minimum Mark: 9.5 / 20
Full details of assessment and respective weight	Exam: 2h, + project, PBL, etc.		
Learning Method(s) and Delivery Mode(s)	According to D.2.5 and D.2.4		
Main Aim	The aim of the "Sustainable Agriculture" discipline is to provide students with an understanding of agricultural practices that promote environmental, social, and economic sustainability. The course focuses on methods that conserve natural resources, reduce environmental impacts, and improve food security while supporting the livelihoods of farmers and agricultural communities.		
Objectives	• To explore the core principles of sustainable agriculture, including soil health, biodiversity, and ecosystem management. • To study and implement techniques for efficient water use, soil conservation, and nutrient management in farming. • To evaluate the environmental consequences of traditional agricultural practices and explore sustainable alternatives. • To understand the social and economic factors that influence sustainable agriculture and the well-being of farming communities. • To design and apply sustainable farming systems that balance productivity with environmental stewardship.		
Learning Outcomes	• Explain key concepts and principles of sustainable agriculture and how they differ from conventional agricultural practices. • Identify and describe the environmental impacts of different agricultural methods and the benefits of sustainable alternatives.		

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	• Develop and implement agricultural practices that promote soil health, water conservation, and biodiversity. • Conduct assessments of farming practices to identify opportunities for improvement and sustainability. • Demonstrate a commitment to environmental responsibility and the sustainable management of agricultural resources. • Collaborate effectively with farmers, agribusinesses, and community organizations to promote sustainable practices. • Exhibit problem-solving skills and innovative thinking in addressing the challenges of sustainable agriculture.
Knowledge	• Students will learn about ecological principles related to sustainable agriculture, including nutrient cycling, energy flows, and ecosystem services. • Knowledge of soil properties, soil conservation techniques, and water management practices to optimize crop production sustainably. • Insights into how climate change impacts agriculture and strategies for adaptation and mitigation to ensure food security. • Familiarity with crop rotation, intercropping, agroforestry, organic farming, and integrated pest management (IPM). • Awareness of the importance of biodiversity in agriculture, pollinator health, and methods to protect and restore ecosystems. • Understanding of policies and strategies to promote sustainable agriculture
Skills	• Ability to apply soil conservation methods, manage water resources efficiently, and implement sustainable crop production practices • Competence in assessing environmental impacts of agricultural practices, evaluating soil and water quality, and analysing the efficiency of farming techniques. • Skills in developing and managing sustainable agricultural projects, creating crop management plans, and optimizing resource use. • Ability to use modern technology, such as GIS mapping, precision agriculture tools, and data analysis software, to make informed decisions. • Developing creative solutions to address issues like soil erosion, water scarcity, and pest control while maintaining productivity. • Ability to assess the environmental impact of pesticide and fertilizer use • Skills in planning and managing land use to minimize deforestation

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	• Proficiency in monitoring environmental parameters related to soil, water, and air quality
Transverse / soft competencies	• A strong sense of responsibility for protecting natural resources and commitment to reducing the environmental impact of agriculture. • Ability to work effectively with diverse stakeholders, including farmers, agricultural organizations, policymakers, and community members, to promote sustainable practices. • Willingness to adapt to new agricultural technologies and practices, continuously learn about innovations, and remain flexible in problem-solving. • An ethical mindset that prioritizes food security, social justice, and the equitable distribution of resources in agricultural development. • Effective communication skills for educating and engaging communities on the benefits of sustainable agriculture and influencing positive behavioural change. • Proactive attitude in leading sustainable agricultural projects, advocating for change, and inspiring others to adopt environmentally friendly farming practices. • Ability to critically analyse environmental data and impacts.
Contents	• Soil degradation and erosion • Greenhouse gas emissions from agricultural practices • Water scarcity and management • Impact of pesticides and fertilizers on ecosystems • Deforestation and land use change • Monitorization of parameters • Strategic measure

Code	Title	Number of theoretical contact hours	Number of practical contact hours	Number of individual learning hours	ECTS
C.A.#17/2-3	Sustainable agriculture	28	28	124	6
Compulsory	Soil degradation and erosion	4	4	18	0,9
Domain: Agriculture	Greenhouse gas emissions from agricultural practices	4	4	18	0,9

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Discipline: #17	Water scarcity and management	4	4	18	0,9
Year: 2	Impact of pesticides and fertilizers on ecosystems	4	4	18	0,9
Semester: 3	Deforestation and land use change	4	4	18	0,8
	Monitorization of parameters	4	4	18	0,9
	Strategic measures	4	4	18	0,9
Hours	180	28	28	124	6
ECTSs	6				6,0

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4 Graduate Study Program main structure

The new study program is represented on the table below. It has a duration of 56 weeks, 120 ECTS. Students have 8 optional disciplines indicated in the table by the symbol O, for example discipline O.I.#02.1/1-1 is optional, and 11 compulsory UCs indicated by the letter C, for example discipline C.T.#03/2-3 is compulsory.

Master, 2 years	3600	hrs		Compulsory	11
56 weeks	120	ECTS		Optional	8

Year 1 / Semester 1	900	hrs		Compulsory	2
14 weeks	30	ECTS		Optional	4
Soft Skills for Green transition		O.S.#01.1/1-1	Semestral (S)		
Communication in Greening Process	Optional	O.S.#01.2/1-1			
Communication and Promotion of Green transition		O.S.#01.3/1-1			
	Total	C	Course	Seminar / Lab / Project	AW
Total number of hours	120	42	14	28	78
Number of ECTS	4				
Critical Raw Materials		O.I.#02.1/1-1	Semestral (S)		
Resources Management	Optional	O.I.#02.2/1-1			
Materials in Circular Economy		O.I.#02.3/1-1			
	Total	C	Course	Seminar / Lab / Project	AW
Total number of hours	210	84	28	56	126
Number of ECTS	7				
Sustainable Supply Chain Management	Compulsory	C.T.#03/1-1	Semestral (S)		
	Total	C	Course	Seminar / Lab / Project	AW
Total number of hours	180	56	28	28	124
Number of ECTS	6				
Waste Management	Compulsory	C.W.#04/1-1	Semestral (S)		
	Total	C	Course	Seminar / Lab / Project	AW
Total number of hours	150	42	14	28	108

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Number of ECTS	5				
Ethics and Integrity Research Ethics Ethics in Transition to Green	Optional	O.S.#05.1/1-1 O.S.#05.2/1-1 O.S.#05.3/1-1	Semestral (S)		
	Total	C	Course	Seminar / Lab / Project	AW
Total number of hours	90	28	14	14	62
Number of ECTS	3				
Sustainable Development Sustainable and Green Production Technologies Green Technologies for Fabrication Processes	Optional	O.G.#06.1/1-1 O.G.#06.2/1-1 O.G.#06.3/1-1	Semestral (S)		
	Total	C	Course	Seminar / Lab / Project	AW
Total number of hours	150	56	28	28	94
Number of ECTS	5				

Year 1 / Semester 2 14 weeks	900 30	hrs ECTS		Compulsory Optional	4 2
LCA and Sustainable Rehabilitation of Buildings	Compulsory	C.C.#07/1-2	Semestral (S)		
	Total	C	Course	Seminar / Lab / Project	AW
Total number of hours	90	28	14	14	62
Number of ECTS	3				
Waste Technologies	Compulsory	C.W.#08/1-2	Semestral (S)		
	Total	C	Course	Seminar / Lab / Project	AW
Total number of hours	210	84	28	56	126
Number of ECTS	7				
Sustainable Transportation	Compulsory	C.T.#09/1-2	Semestral (S)		
	Total	C	Course	Seminar / Lab / Project	AW
Total number of hours	210	84	28	56	126
Number of ECTS	7				

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Sustainable Energy Systems and Environmental Impact	Compulsory	C.E.#10/1-2	Semestral (S)		
	Total	C	Course	Seminar / Lab / Project	AW
Total number of hours	210	84	28	56	126
Number of ECTS	7				
Digitalization IT in Green Transition Digital infrastructure for Greening Process	Optional	O.S.#11.1/1-2 O.S.#11.2/1-2 O.S.#11.3/1-2	Semestral (S)		
	Total	C	Course	Seminar / Lab / Project	AW
Total number of hours	90	28	14	14	62
Number of ECTS	3				
Research #1 Industrial Practice #1 Greening Case Studies Analysis #1	Optional	O.G.#12.1/1-2 O.G.#12.2/1-2 O.G.#12.3/1-2	Semestral (S)		
	Total	C	Course	Seminar / Lab / Project	AW
Total number of hours	90	28	0	28	62
Number of ECTS	3				

Year 2 / Semester 3	900	hrs		Compulsory	4
14 weeks	30	ECTS		Optional	2
Green Buildings (nZEB)	Compulsory	C.C.#13/2-3	Semestral (S)		
	Total	C	Course	Seminar / Lab / Project	AW
Total number of hours	150	42	14	28	108
Number of ECTS	5				
Sustainable residential activities	Compulsory	C.R.#14/2-3	Semestral (S)		
	Total	C	Course	Seminar / Lab / Project	AW
Total number of hours	180	56	28	28	124
Number of ECTS	6				

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Warehousing and Distribution Centers Greening Warehouses Optimization in Warehousing	Optional	O.T.#15.1/2-3 O.T.#15.2/2-3 O.T.#15.3/2-3	Semestral (S)		
	Total	C	Course	Seminar / Lab / Project	AW
Total number of hours Number of ECTS	150 5	42	14	28	108
Climate Change Mitigation and Adaptation in the Energy Sector	Compulsory	C.E.#16/2-3	Semestral (S)		
	Total	C	Course	Seminar / Lab / Project	AW
Total number of hours Number of ECTS	150 5	42	14	28	108
Sustainable agriculture	Compulsory	C.A.#17/2-3	Semestral (S)		
	Total	C	Course	Seminar / Lab / Project	AW
Total number of hours Number of ECTS	180 6	56	28	28	124
Research #2 Industrial Practice #2 Greening Case Studies Analysis #2	Optional	O.G.#18.1/2-3 O.G.#18.2/2-3 O.G.#18.3/2-3	Semestral (S)		
	Total	C	Course	Seminar / Lab / Project	AW
Total number of hours Number of ECTS	90 3	28	0	28	62

Year 2 / Semester 4 14 weeks	900 30	hrs ECTS		Compulsory Optional	1 0
Disertation	Compulsory	C.G.#19/2-4	Semestral (S)		
	Total	C	Course	Seminar / Lab / Project	AW
Total number of hours Number of ECTS	900 30	56	0	56	844

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5 Conclusions

Within D.2.2 and D.2.3 it was possible to set up a new study program that consist of 4 semesters, 56 weeks, and 120 ECTS. The course has 11 compulsory disciplines and 8 optional that can be chosen among 22 available. The 11 compulsory disciplines in the WATERIC field were deeply described, namely with the inclusion of competences and learning outcomes.

It should be notice that all study structure was built considering the master study programs available on the different partners countries.

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Rev number	Date	Modifications proposed by	Modification

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